

Theory of Flow in Aquicludes Adjacent to Slightly Leaky Aquifers

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Abstract. The theory of flow in aquicludes that are adjacent to slightly leaky aquifers is presented for two cases: (a) aquicludes of finite thickness and (b) aquicludes of semi-infinite thickness. By 'slightly leaky' we simply mean that when the aquifer is being pumped at constant rate, the drawdowns in the aquifer will be predicted by the Theis solution, even though some slight movement of water through adjacent confining beds occurs. It is assumed that flow is essentially horizontal in the aquifer and vertical in the aquiclude. On this basis, analytical expressions for drawdown in the aquiclude are developed for each case. The results are presented in the form of type curves that can be used for aquiclude evaluation. The effect of multiple-well fields is considered in each case. (Key words: Ground water; hydrologic systems; aquicludes)

NOMENCLATURE

B ,	leakage factor ($\sqrt{Kbb'/K'}$), L ;
b ,	thickness of aquifer, L ;
b' ,	thickness of aquiclude, L ;
$-Ei(-u)$, $\int_u^\infty e^{-y}/y dy = -0.5772 - \ln u + u - u^2/2 \cdot 2! + u^3/3 \cdot 3! \dots$;	
K ,	permeability of aquifer, L/T ;
K' ,	permeability of aquiclude, L/T ;
Q ,	pumping rate from aquifer, L^3/T ;
r ,	radial distance from pumping well, L ;
S ,	coefficient of storage for aquifer;
S_s' ,	specific coefficient of storage for aquiclude, L^{-1} ;
s ,	drawdown in aquifer, L ;
s' ,	drawdown in aquiclude, L ;
T ,	transmissibility of aquifer, L^2/T ;
t ,	pumping time, T ;
t_D ,	dimensionless time in aquifer ($1/4u$);
t_D' ,	dimensionless time in semi-infinite aquiclude ($1/4u'$);
t_D'' ,	dimensionless time in finite aquiclude ($1/4u''$);
u ,	$r^2/4at$;
u' ,	$z^2/4\alpha't$;
u'' ,	$b'^2/\pi^2\alpha't$;
$W(u)$,	$-Ei(-u)$;
z ,	vertical distance above bottom of aquiclude, L ;
α ,	hydraulic diffusivity of aquifer, L^2/T ;
α' ,	hydraulic diffusivity of aquiclude, L^2/T .

INTRODUCTION

The flow of water through the confining beds that enclose an aquifer may be negligible, in which case the beds are called 'aquicludes,' or

it may be appreciable, in which case they are called 'aquitards' [Marey, 1964]. Our primary interest is in developing an understanding of flow in slightly permeable beds that lie either above or below an aquifer that is being pumped. We assume that the permeability of the confining bed is low enough that one cannot use a conventional pumping test to detect the fact that small amounts of water are moving through these beds. Under these circumstances, it would appear that the term 'aquiclude' is appropriate, and we will therefore use that term throughout this discussion.

In a subsequent paper, we will consider the case of 'aquitards' adjacent to leaky aquifers; i.e., the case where appreciable quantities of water move through the confining bed such that one can detect the influence on the behavior of fluid levels in aquifer observation wells.

We think that more attention must be focused on these so-called 'aquicludes.' No matter how small the permeability of one bed may be relative to its neighbor, the flow of fluids in the whole system cannot be completely understood if part of the system is ignored. An aquiclude above a prolific aquifer may make an inconsequential contribution to the flow system of a pumping well in the local sense, but it may be of major importance in recharging the aquifer in a regional sense.

There are other situations where a description of flow in the aquiclude is also needed if one is to understand the hydrologic behavior of the system under investigation. In those cases where subsidence occurs in unconsolidated clays adjacent to aquifers that are being overpumped, the slow removal of water from the unconsolidated bed is a primary factor in causing the clay beds to contract. There is ample evidence that subsidence can definitely be related to water withdrawal [Poland and Green, 1962; Poland and Davis, 1967].

In connection with the storage of natural gas in aquifers, which is a new technology that has become quite important in the United States, there is the critical problem of making certain that gas does not leak through the overlying aquiclude. The success of the project depends upon a proper evaluation of the magnitude of the permeability of the layers overlying the storage zone. Thus, the gas storage engineer has to know something about the hydrology of aquicludes [Witherspoon et al., 1962; Witherspoon and Neuman, 1967].

The invasion of seawater into the aquifers of coastal basins is another situation where the hydrologic behavior of existing aquicludes should be known. If one aquifer of a multiaquifer system has already been seriously invaded, one may have to rely on an intervening aquiclude to prevent further degrading of other sources of potable water. This has become a widespread problem in the coastal basins of California [Calif. Dept. Water Resources, 1958, 1965]. Preventing pollution where waste disposal is contemplated may also depend upon the effectiveness of existing aquicludes.

In the following discussion, we shall present an analytical approach to the theory of flow in an aquiclude that is associated with slightly leaky aquifers. By 'slightly leaky' we simply mean that when the aquifer is being pumped at constant rate, the drawdowns in the aquifer will be predicted by the Theis solution. The permeability of the aquiclude is presumed to be sufficiently small so that its contribution to the total flow reaching the pumping well is not enough to give the usual evidence of a leaky aquifer.

It is important to know the limits within which our theory holds. From a practical standpoint, there is no lower limit. In other words,

after a few minutes of pumping time, the Theis solution holds at all values of r including the well bore [Witherspoon et al., 1967, p. 8].

The upper limit will depend on the magnitude of error that one is willing to accept in determining the properties or behavior of the aquiclude. For example, suppose one knows the properties of the aquifer system and wishes to predict the drawdowns in the aquiclude with an error of less than 10%. It will be necessary to know that the drawdown data for the aquifer follow either a β curve for 'small' values of time or an r/B curve for 'large' values of time as defined by Hantush [1960, 1964], but that they do not deviate from the Theis curve by more than 10%. The upper limit, in terms of $t_D = \frac{1}{4}u$, would be reached when this deviation is 10%.

One can easily show that the permeability of an aquiclude can be significant and still meet the criterion of being part of a slightly leaky system. For example, if $r/B \leq 0.01$ in the development by Hantush [1960], the Theis solution holds everywhere in the aquifer except at large values of time. Suppose that we consider the case where $r = 200$ feet, $b = 100$ feet, and $b' = 400$ feet. Since $B = \sqrt{Kbb'/K'}$, one has immediately that $K' = 0.0001 K$ when $r/B = 0.01$. If the aquifer permeability is of the order of 2000 gal/ft² day (~ 100 darcies), the aquiclude permeability would be 0.2 gal/ft²/day (~ 0.01 darcies), which is certainly small but still significant.

AQUICLUDE OF FINITE THICKNESS

We first consider the case where an aquiclude of permeability K' and finite thickness b' is overlain and underlain by slightly leaky aquifers, each of which is being pumped at constant rates Q_1 and Q_2 (Figure 1). Each aquifer is assumed to be homogeneous, isotropic, and of infinite radial extent. The hydraulic diffusivities, however, are not necessarily the same, and we therefore set $\alpha_A = T_A/S_A$ in Aquifer A and $\alpha_B = T_B/S_B$ in Aquifer B.

The flow field within the aquiclude is, of course, three-dimensional, but mathematical analyses of such complex systems are very difficult. However, where there is a large contrast between the permeabilities of aquiclude and aquifer, it can be shown that the horizontal components of flow in the aquiclude are negligibly

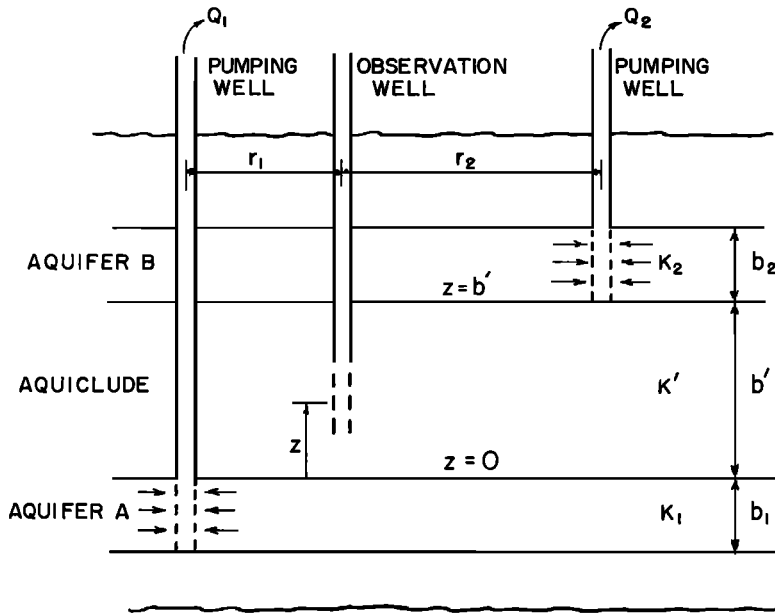


Fig. 1. Diagrammatic cross section of an aquiclude enclosed by two aquifers.

small compared with the vertical components. Neuman [1966] has examined this relationship using a mathematical model where flow was unrestricted in both the aquifer and aquiclude. Except in the vicinity of the pumping well, where small deviations may occur, it was found that the direction of flow in the aquiclude was essentially vertical.

On this basis, we decided to neglect the horizontal components of flow and to consider the problem as being only one-dimensional. Hantush and Jacob [1955] have also made this assumption in their analysis of leaky aquifers. The answers given below, however, will be in terms of $s'(r, z, t)$, because the boundary conditions to be used are dependent on radial distance. We also assumed a confined system of infinite radial extent that remains saturated at all times.

Since we are interested in determining $s'(r, z, t)$, the drawdown in the aquiclude, the boundary value problem for a homogeneous and isotropic system under the above conditions may be expressed as

$$\partial^2 s' / \partial z^2 = (1/\alpha')(\partial s' / \partial t) \quad (1)$$

$$s'(r, z, 0) = 0 \quad (2)$$

$$s'(r, 0, t) = Q_1 / (4\pi T_A) [-Ei(-u_1)] \quad (3)$$

$$s'(r, b', t) = Q_2 / (4\pi T_B) [-Ei(-u_2)] \quad (4)$$

The derivation of (1) has been given by Hantush [1964, p. 328] and was previously used by him in developing a modification of the theory of leaky aquifers [Hantush, 1960].

If boundary conditions 3 and 4 are given by the functions $\varphi_1(t)$ and $\varphi_2(t)$, respectively, Carslaw and Jaeger [1959, p. 103] have shown that the solution to (1) for the conduction of heat under comparable conditions may be obtained using Duhamel's theorem. Their solution may be expressed as

$$s'(z, t) = \frac{2\alpha'\pi}{b'^2} \sum_{n=1}^{\infty} n \sin \frac{n\pi z}{b'} \cdot \int_0^t [\varphi_1(\lambda) - (-1)^n \varphi_2(\lambda)] \cdot \exp \left[-\frac{n^2 \alpha' \pi^2 (t - \lambda)}{b'^2} \right] d\lambda \quad (5)$$

If we define a new variable

$$y^2 = [n^2 \alpha' \pi^2 (t - \lambda)] / b'^2 \quad (6)$$

and introduce the term

$$u'' = b'^2 / (\pi^2 \alpha' t) \quad (7)$$

we can rewrite (5) in the form

$$s'(z, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi z}{b'} \cdot \int_0^{n/\sqrt{u''}} \left[\varphi_1 \left(t - \frac{y^2 b'^2}{n^2 \alpha' \pi^2} \right) - (-1)^n \varphi_2 \left(t - \frac{y^2 b'^2}{n^2 \alpha' \pi^2} \right) \right] y e^{-y^2} dy$$

which can also be written

$$s'(z, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi z}{b'} \cdot \int_0^{n/\sqrt{u''}} \varphi_1 \left(t - \frac{y^2 b'^2}{n^2 \alpha' \pi^2} \right) y e^{-y^2} dy + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi(b' - z)}{b'} \cdot \int_0^{n/\sqrt{u''}} \varphi_2 \left(t - \frac{y^2 b'^2}{n^2 \alpha' \pi^2} \right) y e^{-y^2} dy \quad (8)$$

We are now in a position to return to the original problem described by boundary conditions 3 and 4. We let

$$\varphi_1(t) = Q_1/(4\pi T_A)[-Ei(-u_1)] \quad (9)$$

and

$$\varphi_2(t) = Q_2/(4\pi T_B)[-Ei(-u_2)] \quad (10)$$

Substituting (9) and (10) into equation 8, we have the final solution

$$s'(r, z, t) = \frac{Q_1}{4\pi T_A} W(u_1, u_1'', z/b') + \frac{Q_2}{4\pi T_B} W(u_2, u_2'', 1 - z/b') \quad (11)$$

where

$$W(u_1, u_1'', z/b') = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi z}{b'} \cdot \int_0^{n/\sqrt{u_1''}} -Ei \left[-\frac{n^2 u_1}{n^2 - u_1'' y^2} \right] y e^{-y^2} dy \quad (12)$$

and

$$W(u_2, u_2'', 1 - z/b') = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi(b' - z)}{b'} \cdot \int_0^{n/\sqrt{u_2''}} -Ei \left[-\frac{n^2 u_2}{n^2 - u_2'' y^2} \right] y e^{-y^2} dy \quad (13)$$

The first term on the right side of (11) represents the effect due to the well pumping in the lower aquifer, and the second term represents the effect due to the well in the upper aquifer. Since the two wells do not necessarily start pumping at the same time, we have subscripted u'' in (11) to allow for this. This situation is discussed in more detail below in connection with the generalized solution (see equations 22 and 24).

Since y is now the dummy variable of integration in (12), it must take on all values between the limits of integration. Therefore, for the limiting case, when u_1'' becomes sufficiently small, which means large values of pumping time, the exponential integral term in (12) approaches $-Ei(-u_1)$. The remaining integral approaches $1/2$, and thus 12 becomes

$$W(u_1, u_1'', z/b') = -Ei(-u_1) \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi z}{b'} \quad (14)$$

Since the inverse Fourier sine transform of $1/n$ in the above series is simply $1 - z/b'$ [Churchill, 1958, pp. 289-290], equation 14 can be simplified to

$$W(u_1, u_1'', z/b') = -Ei(-u_1)(1 - z/b') \quad (15)$$

Analogously, if u_2'' becomes sufficiently small, (13) is also simplified to

$$W(u_2, u_2'', 1 - z/b') = -Ei(-u_2)z/b' \quad (16)$$

For the special case where only one well is pumping in the lower aquifer and u_1'' is sufficiently small, (11) becomes

$$s'(r, z, t) = Q_1/(4\pi T_A)[-Ei(-u_1)](1 - z/b') \quad (17)$$

Thus, the ratio between the drawdown in the aquiclude at some given distance r and the drawdown in the aquifer at the same radial distance from the pumping well and at the same pumping time is simply

$$s'/s = 1 - z/b' \quad (18)$$

Since we have assumed no drawdown in the upper aquifer, there will be a linear variation in the drawdown across the aquiclude as the pseudo-steady state is approached. The same results are obtained if the situation is reversed

and Aquifer B is pumped at constant rate instead of Aquifer A. In this case, 18 will become

$$s'/s = z/b' \quad (19)$$

For the case of multiple well systems in either or both aquifers, equation 11 can be generalized as follows. Suppose that there are M wells in the lower aquifer and N wells in the upper aquifer. The general solution can then be written as

$$s'(r, z, t) = \sum_{i=1}^M \frac{Q_i}{4\pi T_A} W(u_i, u_i'', z/b') + \sum_{j=1}^N \frac{Q_j}{4\pi T_B} W(u_j, u_j'', 1 - z/b') \quad (20)$$

Since the wells are likely to start pumping at different times, let t_i represent the time when the i th well in the lower aquifer starts, and t_j represent the time when the j th well in the upper aquifer starts operating. Then equation 20 holds, if for the lower aquifer we let

$$W(u_i, u_i'', z/b') = 0 \quad t \leq t_i \quad (21)$$

and define u_i and u_i'' as

$$u_i = r_i^2/[4\alpha_A(t - t_i)] \quad t > t_i \quad (22)$$

$$u_i'' = b'^2/[\pi^2\alpha'(t - t_i)]$$

and for the upper aquifer, we let

$$W(u_j, u_j'', 1 - z/b') = 0 \quad t \leq t_j \quad (23)$$

and define u_j and u_j'' as

$$u_j = r_j^2/[4\alpha_B(t - t_j)] \quad t > t_j \quad (24)$$

$$u_j'' = b'^2/[\pi^2\alpha'(t - t_j)]$$

If any well is shut down during the operations, the effect is the same as if that well were to continue pumping and an identical recharge well were to start operating at the same location at the time of the shutdown.

AQUICLUDE OF INFINITE THICKNESS

We next consider the case where the aquiclude is so thick that for mathematical purposes it can be considered as being semi-infinite. We shall derive a solution for the situation where Aquifer A is being pumped; the results, of course, apply whether the aquiclude lies above or below the slightly leaky aquifer.

The boundary value problem for these conditions may be expressed as

$$\partial^2 s'/\partial z^2 = (1/\alpha') (\partial s'/\partial t) \quad (1)$$

$$s'(r, z, 0) = 0 \quad (2)$$

$$s'(r, 0, t) = Q_1/(4\pi T_A)[-Ei(-u_1)] \quad (3)$$

$$s'(r, \infty, t) = 0 \quad (25)$$

If we again replace boundary condition 3 by $\varphi_1(t)$, Carslaw and Jaeger [1959, p. 63] have shown that the solution to (1) may be obtained using Laplace transforms and Duhamel's theorem. Their solution may be expressed as

$$s'(z, t) = \int_0^t \varphi_1(\lambda) \frac{z}{2\sqrt{\pi(t-\lambda)\alpha'}} \cdot \exp\left[-\frac{z^2}{4(t-\lambda)\alpha'}\right] d\lambda \quad (26)$$

If we define a new variable

$$y^2 = z^2/[4(t-\lambda)\alpha'] \quad (27)$$

and introduce the term

$$u' = z^2/(4\alpha't) \quad (28)$$

we can rewrite 26 in the form

$$s'(z, t) = \frac{2}{\sqrt{\pi}} \int_{\sqrt{u'}}^{\infty} \varphi_1\left(t - \frac{z^2}{4y^2\alpha'}\right) e^{-y^2} dy \quad (29)$$

We now let

$$\varphi_1(t) = Q_1/(4\pi T_A)[-Ei(-u_1)] \quad (9)$$

and substituting into (29), we have the final solution

$$s'(r, z, t) = Q_1/(4\pi T_A) W(u_1, u') \quad (30)$$

where

$$W(u_1, u') = \frac{2}{\sqrt{\pi}} \int_{\sqrt{u'}}^{\infty} -Ei\left(-\frac{u_1 y^2}{y^2 - u'}\right) e^{-y^2} dy \quad (31)$$

In the same manner as in the case of the finite aquiclude, the effect of a multiple well field can be obtained from

$$s'(r, z, t) = \sum_{i=1}^M Q_i/(4\pi T_A) W(u_i, u_i') \quad (32)$$

where

$$W(u_i, u_i') = 0 \quad t \leq t_i \quad (33)$$

and

$$\begin{aligned} u_i &= r_i^2/[4\alpha_A(t - t_i)] \\ u_i' &= z^2/[4\alpha'(t - t_i)] \end{aligned} \quad t > t_i \quad (34)$$

For sufficiently large values of time, the exponential integral term in (31) approaches $-Ei(-u_i)$, such that

$$W(u_i, u') = -Ei(-u_i)2/\sqrt{\pi} \int_0^\infty e^{-y^2} dy$$

$$\begin{aligned} &= -Ei(-u_i) \operatorname{erf}(\infty) \\ &= -Ei(-u_i) \end{aligned}$$

and (30) reduces to the Theis solution

$$s'(r, z, t) = Q_1/(4\pi T_A)[-Ei(-u_i)] \quad (35)$$

Thus, for sufficiently large values of time, the drawdown in the aquiclude at some given point must become the same as the drawdown in the aquifer at the same radial distance from the pumping well and at the same pumping time. In other words

$$s'/s = 1. \quad (36)$$

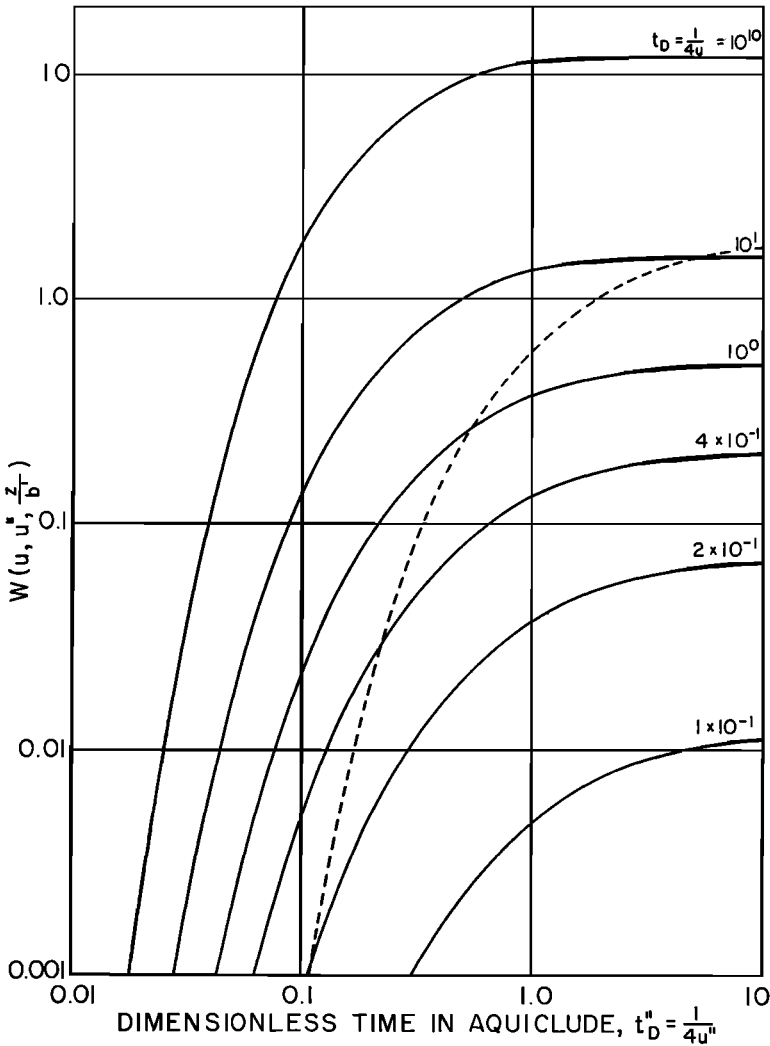


Fig. 2. $W(u, u', z/b')$ versus t_D'' for finite aquiclude ($z/b' = 0.5$).

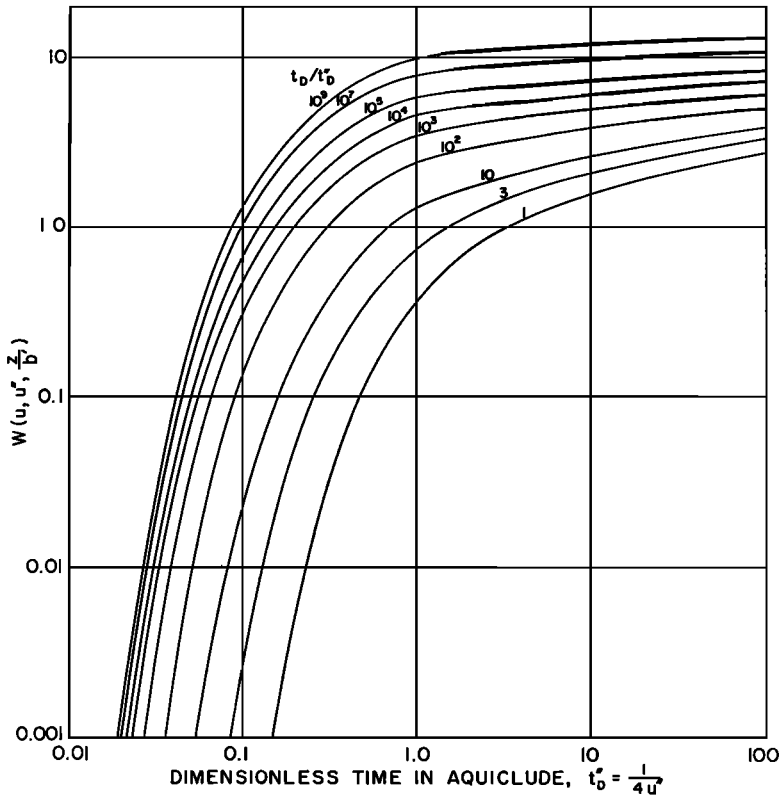


Fig. 3. Type curves for finite aquiclude ($z/b' = 0.5$).

This, of course, could have been anticipated from (18), where b' is taken to be very large ($b' \rightarrow \infty$).

DISCUSSION OF RESULTS

To examine the nature of these solutions, we have used numerical techniques (Adams-Moulton method) to evaluate the functions $W(u, u'', z/b')$ and $W(u, u')$. These functions have been evaluated in terms of t_D , dimensionless time in the aquifer, which is defined as

$$t_D = 1/4u = Tt/Sr^2 \quad (37)$$

t_D' , dimensionless time in the semi-infinite aquiclude, which is defined as

$$t_D' = 1/(4u') = (K't)/(S_s'z^2) \quad (38)$$

and t_D'' , dimensionless time in the finite aquiclude, which is defined as

$$t_D'' = 1/(4u'') = (\pi^2 K't)/(4S_s'b'^2) \quad (39)$$

Extensive tables of results for both $W(u, u'', z/b')$ and $W(u, u')$ have been published elsewhere [Witherspoon *et al.*, 1967]. These results cover a practical range of dimensionless times and values of z/b' (or its complement $1 - z/b'$) ranging from 0 to 0.9.

Figure 2 shows how $W(u, u'', z/b')$ at $z/b' = 0.5$ varies with t_D'' for various values of t_D . For a given situation where there is only one pumping well, it must be kept in mind that both t_D and t_D'' will increase with time in the same manner. For example, the dashed line on Figure 2 shows one of an infinite number of possibilities how $W(u, u'', z/b')$ would increase with time at a given point (r, z) in the aquiclude. Figure 3 shows the same data as that of Figure 2 but in the form of type curves that have been plotted over a practical range of values for the ratio t_D/t_D'' . This ratio will, of course, remain constant for a given pumping test.

In the case of the semi-infinite aquiclude, one will obtain a spread in the curves of $W(u, u')$ similar to that on Figure 3. This is shown on

Figure 4, where the type curves have been plotted in terms of t_D/t_D' .

When there is only one pumping well, the results given by equation 11 should, of course, be identical with those given by equation 30 for appropriately small values of time. This can be demonstrated by converting values of $W(u, u', z/b')$ in terms of t_D'' into their equivalent values in terms of t_D' , using

$$t_D' = (2b'/\pi z)^2 t_D'' \quad (40)$$

Figure 5 shows how results in terms of s'/s obtained from these two equations compare for one value of dimensionless time in the aquifer at $t_D = 0.2$. Essentially the same relationships for different values of z/b' were obtained at all other values of t_D . It can be seen on Figure 5 that for a given value of z/b' , the curve is approaching $1 - z/b'$ as a limit, as predicted by equation 18. In the case of the semi-infinite aquiclude, $z/b' = 0$, and the curve approaches unity in accordance with equation 36.

The results on Figure 5 indicate that one will be able to evaluate many field situations by assuming the semi-infinite case. Although the family of curves shown on Figure 4 does not yield a unique solution between s' and t_D' , we have developed simplified methods of evaluating aquiclude permeabilities under field conditions that give satisfactory results [Witherspoon and Neuman, 1967].

Flow within the aquiclude can be obtained from Darcy's law by using the gradient of the drawdown $\partial s'/\partial z$. From (12) we have

$$\frac{\partial W(u, u', z/b')}{\partial z} = \frac{4}{b'} \sum_{n=1}^{\infty} \cos \frac{n\pi z}{b'} \cdot \int_0^{n/\sqrt{u''}} -Ei \left[-\frac{n^2 u}{n^2 - u'' y^2} \right] y e^{-y^2} dy \quad (41)$$

and the gradient for any situation can be obtained by combining equations 20 and 41. By

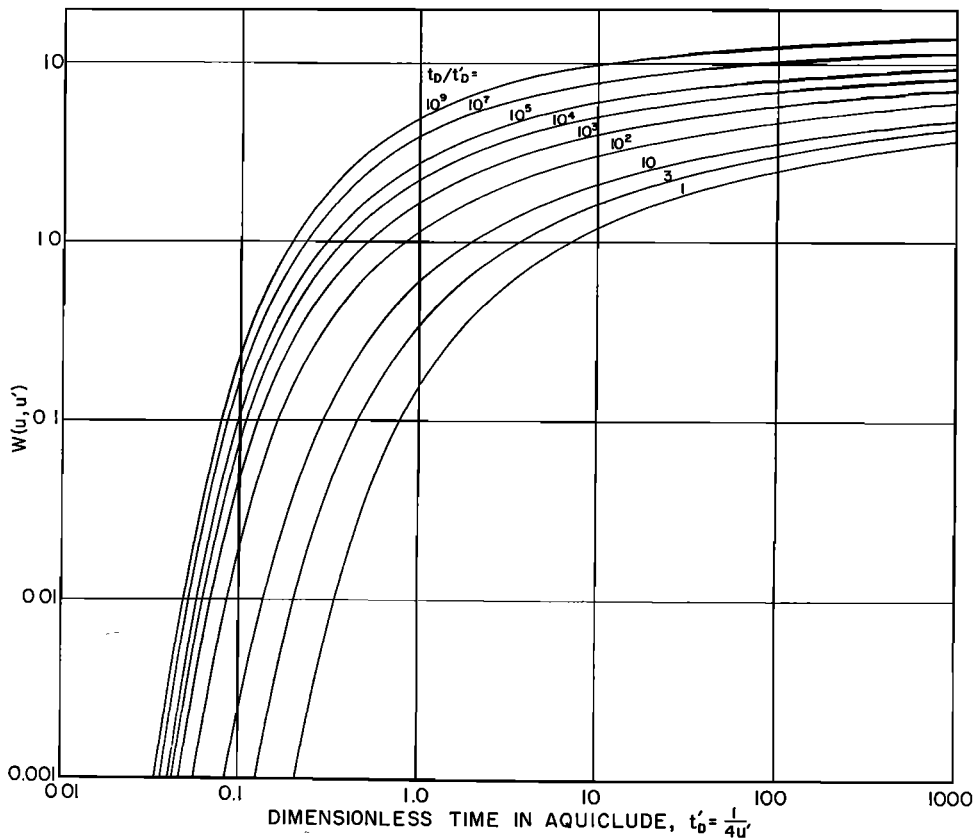


Fig. 4. Type curves for semi-infinite aquiclude.

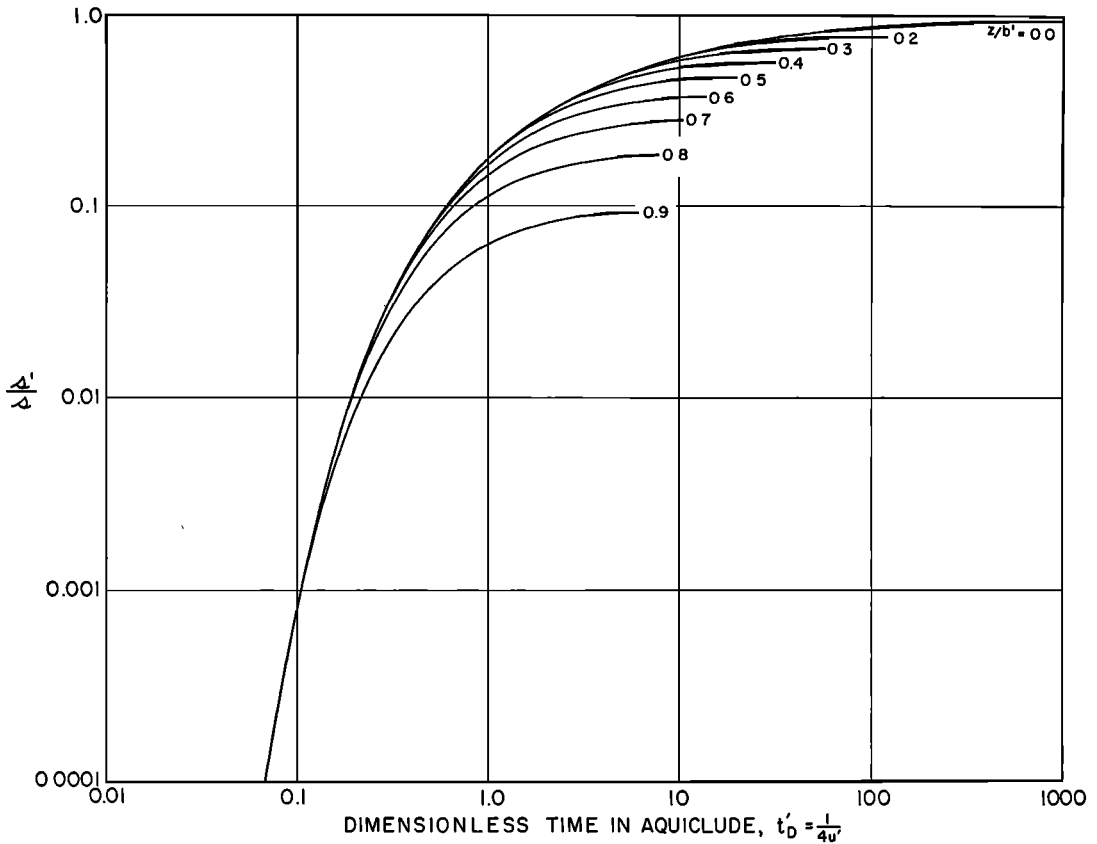


Fig. 5. s'/s versus t'_D at $t_D = 0.2$ for semi-infinite aquiclude compared with results for finite aquiclude at various values of z/b' (after Witherspoon *et al.*, 1967, courtesy AGA).

calculating this gradient at the upper and lower boundaries of the aquiclude, one can determine the total amount of water leaving the aquiclude.

Since it is assumed that subsidence is directly related to changes in pore pressure within the aquiclude and the total amount of water leaving the aquiclude, one should now be able to predict how subsidence will occur as a function of time. Since the phenomenon appears to be only slightly reversible [Poland and Davis, 1967], one should also be able to predict the cumulative effect of cyclic pumping.

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